

5C34 : ADVANCED AI

An Introduction to CLIP: Unifying Text and Images via Contrastive Learning

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Part 1: Introduction (The Problem Context)

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The Problem Context

An Introduction to CLIP: Unifying Text and Images via Contrastive Learning

Primary Papers:

- Learning Transferable Visual Models From Natural Language Supervision (Radford et al., OpenAI, 2021)
- Reproducible Scaling Laws for Contrastive Language-Image Learning (Cherti et al., CVPR 2023)

The Bottleneck in Standard Computer Vision

The “Closed-Set” Paradigm (Handout 06 Deep Learning)

- Standard models (ResNet, VGG, AlexNet) trained on fixed N classes using Softmax + Cross-Entropy loss

Technical Limitations:

- Labor-intensive: Requires massive, manually curated datasets (e.g., ImageNet with crowdsourced labels)
- Rigid: Cannot classify categories outside the fixed N classes without architectural changes and retraining
- Weak generalization: Models learn spurious correlations rather than true semantic concepts

Inspiration from NLP (Handouts 09 & 11 Deep Learning)

- Modern NLP scaled by using raw, unlabelled internet text (self-supervised learning)

The Key Proposition:

- Replace human-annotated one-hot labels with raw textual descriptions
- Leverage naturally occurring image-text pairs from the web (e.g., alt-text, captions)
- Training set: 400 million (Image, Text) pairs scraped from the internet

Part 2: Main Paper 1

Paper Focus: “Learning Transferable Visual Models From Natural Language Supervision”
(Radford et al., OpenAI, 2021)

The Dual-Encoder Architecture

Two Separate Encoders:

- **Text Encoder:** Masked Self-Attention Transformer (Handout 09)
Processes tokenized text $\rightarrow$ dense vector embedding
- **Image Encoder:** ResNet-50 (Handout 06) or Vision Transformer / ViT (Week 04)
Processes image $\rightarrow$ dense vector embedding

Linear Projection: Both encoders map to a shared latent space of dimension d

Key Insight: Images and text must “speak the same mathematical language”

The Contrastive Pre-Training Objective (InfoNCE Loss)

InfoNCE Loss (Contrastive Learning):

- Instead of predicting exact text (too sparse), predict which text matches which image
- Given batch of N (Image, Text) pairs:
 - Compute cosine similarity between all $N \times N$ pairings
 - Maximize similarity of N correct pairs (diagonal)
 - Minimize similarity of $N^2 - N$ incorrect pairs (off-diagonal)

(1) Contrastive pre-training

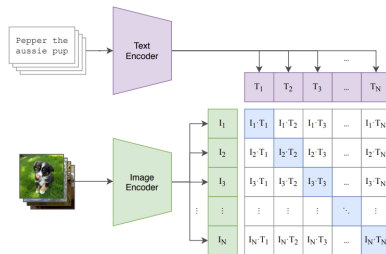


Figure 1. Contrastive Pre-Training

Numpy Pseudocode (Technical Deep-Dive)

The Training Loop

- Extract features

$I_f = \text{image_encoder}(I), \quad T_f = \text{text_encoder}(T)$

- Project and normalize

$I_e = \text{l2_normalize}(W_i I_f), \quad T_e = \text{l2_normalize}(W_t T_f)$

- Compute logits

$$\text{logits} = I_e T_e^T \cdot \exp(\tau)$$

where τ is a learnable temperature parameter

- Symmetric Cross-Entropy Loss

$\text{loss}_i = \text{CrossEntropy}(\text{logits}, \text{labels}, \text{axis} = 0)$

$\text{loss}_t = \text{CrossEntropy}(\text{logits}, \text{labels}, \text{axis} = 1)$

$$\text{loss} = \frac{\text{loss}_i + \text{loss}_t}{2}$$

```
# image_encoder - ResNet or Vision Transformer
# text_encoder - CBOW or Text Transformer
# I[n, h, w, c] - minibatch of aligned images
# T[n, l] - minibatch of aligned texts
# W_i[d_i, d_e] - learned proj of image to embed
# W_t[d_t, d_e] - learned proj of text to embed
# t - learned temperature parameter

# extract feature representations of each modality
I_f = image_encoder(I) #[n, d_i]
T_f = text_encoder(T) #[n, d_t]

# joint multimodal embedding [n, d_e]
I_e = l2_normalize(np.dot(I_f, W_i), axis=1)
T_e = l2_normalize(np.dot(T_f, W_t), axis=1)

# scaled pairwise cosine similarities [n, n]
logits = np.dot(I_e, T_e.T) * np.exp(t)

# symmetric loss function
labels = np.arange(n)
loss_i = cross_entropy_loss(logits, labels, axis=0)
loss_t = cross_entropy_loss(logits, labels, axis=1)
loss = (loss_i + loss_t)/2
```

Figure 3 from Radford et al. (2021), CLIP paper, page 5

Zero-Shot Inference Mechanism

No Linear Classifier Head Required Inference Process:

- Pass target image through Image Encoder $\rightarrow$ Image embedding vector
- Prompt Engineering: Embed class names into templates:
 - “A photo of a {dog}”
 - “A photo of a {cat}”
 - “A photo of a {plane}”
- Pass all text prompts through Text Encoder $\rightarrow$ Text embedding vectors
- Compute cosine similarity between image vector and all text vectors
- Apply Softmax $\rightarrow$ Highest probability = predicted class

(2) Create dataset classifier from label text

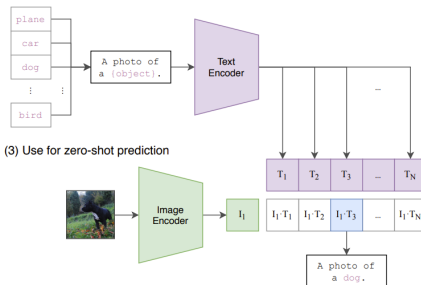


Figure 3. Zero-Shot Inference

Key Findings from Radford et al.

Zero-Shot Performance:

- CLIP matched fully supervised ResNet-50 on ImageNet: 76.2% accuracy
- Achieved without ever training on ImageNet labels

Robustness to Distribution Shift:

- Standard ImageNet-trained models collapse on shifted datasets
- CLIP maintains stable performance across:
 - ImageNet-V2
 - ImageNet-Sketch
 - ImageNet-R
 - ObjectNet
- Why? Learned semantic concepts through language, not pixel-level memorization

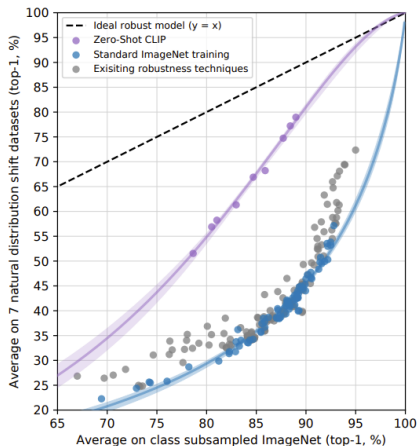


Figure 4. Robustness to Distribution Shift

Part 3: Main Paper 2

Paper Focus: “Reproducible Scaling Laws for Contrastive Language-Image Learning”
(Cherti et al., CVPR 2023)

The Reproducibility Problem

The Closed-Data Problem:

- OpenAI kept their 400M image-text dataset proprietary
- No way for researchers to verify or reproduce results

OpenCLIP Solution (Cherti et al., 2023):

- Trained on open-source LAION-400M and LAION-2B datasets
- Fully reproducible architecture and training code released
- Goal: Establish whether CLIP's success generalizes to open data

Establishing Scaling Laws

What are Scaling Laws?

- Empirical observation: performance improves predictably as a power-law function of:
 - Model size
 - Dataset size
 - Compute budget

Experimental Setup

- Trained models from ViT-B/32 to ViT-G/14
- Dataset sizes: 400M to 2B image-text pairs

Key Finding

- Contrastive language-image learning obeys strict scaling laws
- More compute + more data → reliably improved zero-shot accuracy

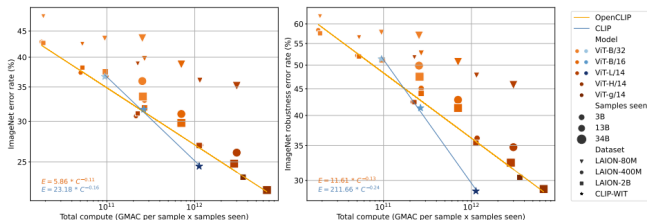


Figure 5. Scaling Laws of Contrastive Language-Image Learning

Pushing the State-of-the-Art

Results:

- OpenCLIP ViT-G/14 trained on LAION-2B: 80.1% zero-shot accuracy on ImageNet
- Surpassed original OpenAI CLIP
- Proved open-source models can match/exceed proprietary systems

Implications:

- Democratizes access to foundation models
- Enables broader research community participation

Part 4: Conclusions & Impact

CLIP as a “Foundation Model”

Beyond Classification:

- Image retrieval, visual question answering, object detection

Link to Generative Models (Handout 10):

- Text-to-image diffusion models (DALL-E 2, Stable Diffusion, Midjourney)
- Use frozen CLIP text encoder to condition image generation
- CLIP embedding tells the diffusion model what the prompt “means” visually

Foundation Model: Pre-trained backbone for countless downstream applications

Summary & Final Thoughts

Technical Takeaways:

- Contrastive learning on noisy image-text pairs $>$ supervised learning on clean labels
- Mapping multiple modalities to shared latent space enables zero-shot transfer
- Architecture follows predictable scaling laws

Limitations & Future Work:

- Struggles with fine-grained counting and text-heavy images
- Video-language contrastive learning (emerging research area)
- Improving robustness to adversarial examples

Thank you!

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